

Electrodynamics (PHY303) - Midsemester Examination 1 2025

Instructions:

- Please write short and precise answers. If you make an assumption or an approximation, please write it clearly.
- Please cross any rough work or wrong answer. Illegible answers will not be graded.
- Calculators (if needed) are allowed and encouraged. Sharing a calculator is not allowed.
- You can use one hand written formula sheet (A4 size, one side).

Maximum Marks=25, Time allowed=60 Minutes

1. Find the electric field inside a sphere which carries a charge density proportional to the distance from the origin (which is at its centre) $\rho = \alpha r$, for some constant α . [6]
2. A flat non-conducting sheet lies in the xy plane. The only charges in the system are on this sheet. In the half space above sheet $z > 0$ the potential is $\phi = \phi_0 e^{-kz} \cos kx$ where ϕ_0, k are constants.
 - (a) Find the electric field.
 - (b) Draw the electric field lines in the $x - z$ plane.
 - (c) Compute the surface charge density on the sheet. [2,1,3]
3. Find the work done to remove the charge q against the force of a grounded conducting sphere of radius a given by the equation [6]

$$|\mathbf{F}| = \frac{1}{4\pi\epsilon_0} \frac{q^2}{a^2} \left(\frac{a}{y}\right)^3 \left(1 - \frac{a^2}{y^2}\right)^{-2}$$

4. Two charges $+q$ are placed at a distance $2d$ from each other.
 - (a) Compute force between the two point charges.
 - (b) Now, a grounded, conducting sphere is placed midway between them. Find the positions of the image charge(s) appropriately and find their charge.
 - (c) Find the approximate minimum (leading order) radius a which will neutralise the force. [1,2,4]

MP25007

PHY 303- Electrodynamics Mid Semester Examination II

Please write short and concise answers. Illegible answers will not be graded.

Maximum marks =25, Time = 60 minutes

1. Expand the following function in terms of Legendre Polynomials and write down the first two terms [5]

$$f(x) = \begin{cases} +1 & \text{for } x > 0 \\ -1 & \text{for } x < 0 \end{cases}$$

2. The lowest order term in the vector potential due to a current carrying loop in the xy plane is given by

$$A_\varphi = \frac{\mu_0 I a^2 r \sin \theta}{4(a^2 + r^2)^{3/2}}$$

Find the components of the magnetic field. [5]

3. A long, solid dielectric cylinder of radius a is permanently polarized so that the polarization is radially outward everywhere, with a magnitude proportional to the distance from the axis of the cylinder, i.e.,

$$\vec{P} = \frac{1}{2} P_0 r \hat{r}$$

Find the volume charge density in the cylinder. [5]

4. A charge q is placed at a distance d from a dielectric sphere of radius a with permittivity ϵ . Assume that the line joining the centre of the sphere to the charge is along the z-axis. An image charge iq is at $z = a^2/d$.

- Write the expression for the electrostatic potential outside the sphere and expand the potential for the regions $r < d$ and $r > d$ in terms of Legendre polynomials.
- Put an image charge q'' at $z = d$ and write down the expression for the potential inside the sphere, and expand in terms of Legendre polynomials (Hint: Remember to use the correct permittivity in all the cases).
- Using the boundary conditions

$$(\vec{D}_2 - \vec{D}_1) \cdot \hat{n} = \sigma, \quad (\vec{E}_2 - \vec{E}_1) \times \hat{n} = 0,$$

find the charges q'' and q' in terms of q (Hint: Write down the boundary conditions in terms of derivatives of Φ). [3+3+4]

Useful Formulae:

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn}$$

Divergence in cylindrical and spherical coordinates:

$$\frac{1}{\rho} \frac{\partial (\rho A_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z}$$

$$\frac{1}{r^2} \frac{\partial (r^2 A_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\varphi}{\partial \varphi}$$

Curl in cylindrical and spherical coordinates:

$$\left(\frac{1}{\rho} \frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z} \right) \hat{\rho} + \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) \hat{\varphi} + \frac{1}{\rho} \left(\frac{\partial (\rho A_\varphi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \varphi} \right) \hat{z}$$

$$\frac{1}{r \sin \theta} \left(\frac{\partial}{\partial \theta} (A_\varphi \sin \theta) - \frac{\partial A_\theta}{\partial \varphi} \right) \hat{r} + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \varphi} - \frac{\partial}{\partial r} (r A_\varphi) \right) \hat{\theta} + \frac{1}{r} \left(\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right) \hat{\varphi}$$

PHY 303- Electrodynamics-End Semester Examination

Please write short and concise answers. Illegible answers will not be graded.
Maximum marks =50, Time = 3 hours

1. (a) Two uniform infinite sheets of surface charge densities $+\sigma$ and $-\sigma$ intersect at right angles. Find the magnitude and direction of the electric field everywhere and draw the electric field lines. [5]
- (b) The space between two concentric shells of radii a and b ($a < b$) is filled with an inhomogeneous material of dielectric constant

$$\epsilon = \frac{\epsilon_0}{1 + Kr}$$

where ϵ_0 and K are constants. A charge q is placed at the inner surface while the outer surface is grounded. Find

- i. The displacement and the electric field in the region $a < r < b$
 - ii. The capacitance of the device.
 - iii. The polarization charge density in the region $a < r < b$. [2,2,1]
2. (a) A current distribution gives rise to the magnetic vector potential

$$\vec{A}(x, y, z) = x^2 y \hat{i} + y^2 x \hat{j} - xyz \hat{k}$$

Find the corresponding magnetic field $\vec{B}$ at $(-1, 2, 5)$. [3]

- (b) The magnetic vector potential due to a small loop carrying current I may be expressed in the form

$$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{Id\vec{l}}{R},$$

where $R = \sqrt{r^2 + r'^2 - 2rr' \cos \theta}$.

Expand the expression in terms of Legendre polynomials at large distance (assuming Coulomb Gauge $\vec{\nabla} \cdot \vec{A} = 0$). [7]

3. (a) In a region of empty space, the magnetic field is given by

$$\vec{B} = B_0 e^{ax} \sin(ky - \omega t) \hat{x}.$$

Calculate $\vec{E}$. [5]

- (b) Given a plane polarised electric wave,

$$\vec{E} = \vec{E}_0 \exp \left\{ i\omega \left[t - \frac{n}{c} (\vec{k} \cdot \vec{r}) \right] \right\}$$

derive from Maxwell equations, relations between $\vec{E}$, $\vec{k}$ and $\vec{H}$ fields. Find the velocity of propagation and the refractive index. [3,1,1]

4. (a) Compute the intensity of the standing electromagnetic wave

[5]

$$E_y(x, t) = 2E_0 \cos kx \cos \omega t$$

$$B_z(x, t) = 2B_0 \sin kx \sin \omega t.$$

- (b) A stationary observer measures the Earth's magnetic field to be $30\mu T$. What will be the magnetic and electric field (if any) measured by an observer moving past with a velocity 250 m/s perpendicular to the direction of the magnetic field (Hint: choose axes appropriately!).

[5]

5. The scalar and vector potential of a dipole are given by

$$\Phi(r, \theta, t) = -\frac{p_0 \omega \cos \theta}{4\pi \epsilon_0 c r} \sin[\omega(t - r/c)]$$

$$\vec{A}(r, \theta, t) = -\frac{\mu_0 p_0 \omega}{4\pi r} \sin[\omega(t - r/c)] \hat{z} \rightsquigarrow \hat{z}, \hat{\theta}, \hat{\phi}$$

Compute

- (a) The electric and magnetic fields (assuming Lorenz gauge).
 (b) The Poynting vector.
 (c) Intensity and the time averaged power radiated.

[4,2,4]

Useful Formulae

Legendre Polynomials:

$$P_0(x) = 0$$

$$P_1(x) = x$$

$$P_2(x) = \frac{1}{2}(3x^2 - 1)$$

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn}$$

Divergence and Curl in spherical coordinates:

$$\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\varphi}{\partial \varphi}$$

$$\begin{aligned} & \frac{1}{r \sin \theta} \left(\frac{\partial}{\partial \theta} (A_\varphi \sin \theta) - \frac{\partial A_\theta}{\partial \varphi} \right) \hat{r} \\ & + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \varphi} - \frac{\partial}{\partial r} (r A_\varphi) \right) \hat{\theta} \\ & + \frac{1}{r} \left(\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right) \hat{\varphi} \end{aligned}$$

Electromagnetic field tensor:

$$F^{ij} = \begin{bmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & B_x \\ E_z/c & -B_y & B_x & 0 \end{bmatrix}$$